

Customer Segmentation Using RFM Analysis

Business Problem

The retail business has a large volume of historical transaction data but lacks a structured approach to understand differences in customer purchasing behavior and customer value.

Without identifying high-value, loyal, emerging, inactive, and at-risk customers, the business may struggle to focus retention and customer engagement efforts on the customer groups that contribute most to revenue.

This project analyzes historical retail transaction data using Recency, Frequency, and Monetary (RFM) analysis to measure customer purchasing behavior, segment customers based on their value and engagement, and generate actionable insights that can support customer retention, loyalty, and targeted marketing decisions.

Business Understanding

Business Objective

The objective of this project is to:

- Analyze customer transaction behavior.
- Quantify customer value using RFM analysis.
- Segment customers according to their purchasing characteristics.
- Identify high-value, loyal, emerging, at-risk, and inactive customers.
- Understand revenue concentration across customer segments.
- Identify opportunities for retention and repeat purchasing.
- Support differentiated customer engagement and marketing strategies.

Business Questions

1. Which customers generate the highest monetary value for the business?
2. Which customers purchase most recently, most frequently, and spend the most?
3. How concentrated is revenue among the highest-value customers?
4. What are the Recency, Frequency, and Monetary characteristics of the customer base?
5. How can customers be scored and compared based on their RFM behavior?
6. How can customers be segmented into meaningful behavioral groups using their RFM scores?
7. Which customer segments contribute the greatest customer value and revenue?
8. Which customers or segments show characteristics associated with inactivity or retention risk?
9. Which customer segments represent opportunities for strengthening loyalty and repeat purchasing?
10. How can customer segments support differentiated retention, loyalty, and engagement strategies?

Data Understanding

Dataset Overview

The project uses the UCI Online Retail Dataset, which contains transactional data from an online retail business.

The original dataset contains 541,909 transaction records and 8 attributes, covering transactions from 1 December 2010 to 9 December 2011.

Dataset Features

Feature	Description
InvoiceNo	Transaction / invoice ID
StockCode	Product ID
Description	Product name
Quantity	Number of items purchased
InvoiceDate	Purchase date and time
UnitPrice	Price per item
CustomerID	Unique customer ID
Country	Customer location

Data Understanding Objectives

The main objectives during data understanding were to:

- Understand the dataset structure and attributes.
- Identify missing values and duplicate records.
- Identify cancelled transactions and product returns.
- Calculate revenue using $\text{Quantity} \times \text{UnitPrice}$.
- Understand customer purchasing behavior.
- Determine whether the dataset supports RFM analysis.
- Prepare the data for customer-level segmentation.

Initial Data Quality Findings

The original dataset contains:

- 541,909 records
- 8 columns
- 135,080 missing CustomerID values
- 1,454 missing Description values
- 5,268 duplicate records
- Cancelled invoices and negative-quantity transactions representing returns/refunds.

Because RFM analysis requires customer-level purchasing information, records without a valid CustomerID cannot be reliably assigned to individual customers.

Data Cleaning & Preparation

The transaction data was cleaned and prepared before performing customer-level analysis.

Data Cleaning

The following steps were performed:

- Removed records with missing CustomerID values.
- Removed duplicate transaction records.
- Excluded cancelled orders and transactions with negative quantities.
- Converted InvoiceDate into datetime format.
- Removed invalid transactions where required for revenue calculation.
- Created a customer-level dataset from the cleaned transaction data.

Revenue Calculation

A new Revenue column was created:

$$\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$$

This metric was then aggregated at the customer level to calculate each customer's total monetary value.

RFM Metrics

Three customer-level metrics were calculated:

Recency

Measures how recently a customer made a purchase.

$$\text{Recency} = \text{Reference Date} - \text{Most Recent Purchase Date}$$

Lower recency values indicate more recent customer activity.

Frequency

Measures how often a customer purchased.

$$\text{Frequency} = \text{Number of Unique Invoices}$$

Higher frequency indicates more frequent purchasing behavior.

Monetary

Measures the total revenue generated by a customer.

$$\text{Monetary} = \text{Sum of Revenue}$$

Higher monetary values indicate higher customer value.

Clean Customer-Level Dataset

After cleaning and aggregation, the analysis contained:

- 4,338 customers
- 18,532 orders
- 8.89M total revenue
- 2,049 average revenue per customer

The final customer-level dataset was therefore suitable for RFM scoring and customer segmentation.

Exploratory Data Analysis

Exploratory Data Analysis was performed to understand customer purchasing patterns, revenue concentration, and customer engagement.

Overall Customer and Revenue Metrics

Metric	Value
Total Customers	4,338
Total Orders	18,532
Total Revenue	8.89M
Average Customer Revenue	2,049
Average Recency	92.54 days
Average Frequency	4.27 orders
Average Monetary Value	2,049

The results show that customer value is not evenly distributed across the customer base. A relatively small group of highly engaged customers contributes a large proportion of total revenue.

Revenue by Country

After cleaning the transaction data, the highest-revenue countries were:

Country	Revenue
United Kingdom	7.29M
Netherlands	285.45K
EIRE	265.26K
Germany	228.68K
France	208.93K

The United Kingdom represents the largest revenue-generating market in the dataset.

Highest-Value Customers

The highest-value customers based on monetary value include:

Customer ID	Revenue
14646	280.21K
18102	259.66K
17450	194.39K
16446	168.47K
14911	143.71K

These customers represent important high-value relationships and should be considered when designing retention and loyalty strategies.

RFM Calculation

RFM analysis was used to convert transactional behavior into measurable customer-level characteristics.

Each customer was evaluated across three dimensions:

RFM Dimension	Business Meaning
Recency	How recently the customer purchased
Frequency	How often the customer purchased
Monetary	How much revenue the customer generated

Customers were scored according to their relative RFM behavior.

The resulting RFM scores allowed customers with different purchasing patterns to be compared consistently.

For example:

- High Recency score → recently active customer
- High Frequency score → frequent purchaser
- High Monetary score → high-value customer

The combined RFM scores were then used to create meaningful customer segments.

Customer Segmentation

Based on the RFM analysis, customers were divided into six behavioral segments:

Segment	Customers	Customer %	Revenue	Revenue %
Champions	942	21.72%	5.74M	64.56%
Loyal Customers	460	10.60%	910.76K	10.25%

At-Risk Customers	661	15.24%	823.91K	9.27%
Potential Loyalists	511	11.78%	508.60K	5.72%
Other	690	15.91%	382.18K	4.30%
Lost Customers	1,074	24.76%	523.80K	5.89%

Segment Characteristics

Champions

Champions are the highest-value and most engaged customer group.

- 942 customers
- 21.72% of the customer base
- 64.56% of total revenue
- Average Recency: 12.50 days
- Average Frequency: 11.20 orders
- Average Monetary: 6,091.24

Loyal Customers

These customers demonstrate relatively consistent purchasing behavior and contribute meaningful revenue.

- 460 customers
- 10.60% of customers
- 10.25% of revenue
- Average Recency: 38.45 days
- Average Frequency: 5.29 orders
- Average Monetary: 1,979.91

At-Risk Customers

These customers have previously generated meaningful value but show reduced recent engagement.

- 661 customers
- 15.24% of customers
- 9.27% of revenue
- Average Recency: 150.64 days
- Average Frequency: 3.41 orders
- Average Monetary: 1,246.46

Potential Loyalists

These customers have purchased recently but have not yet developed high purchasing frequency.

- 511 customers

- 11.78% of customers
- 5.72% of revenue
- Average Recency: 16.41 days
- Average Frequency: 2.12 orders
- Average Monetary: 995.31

Other Customers

This group contains customers with comparatively lower purchasing frequency and monetary contribution.

- 690 customers
- 15.91% of customers
- 4.30% of revenue

Lost Customers

These customers show very low recent engagement and purchasing frequency.

- 1,074 customers
- 24.76% of customers
- 5.89% of revenue
- Average Recency: 216.68 days
- Average Frequency: 1.10 orders
- Average Monetary: 487.71

Visualization & Dashboard

A Power BI dashboard was developed to communicate the RFM analysis and customer segmentation results.

The dashboard includes:

Executive Overview

- Total Customers
- Total Orders
- Total Revenue
- Average Customer Revenue
- Monthly Revenue Trend

RFM Analysis

- Average Recency
- Average Frequency
- Average Monetary Value
- Top customers by revenue
- Top customers by purchase frequency
- RFM Customer Value Matrix

Customer Segmentation

- Customer count by segment
- Revenue by segment
- Revenue contribution by segment
- Average RFM metrics by segment

Retention & Risk Analysis

- At-Risk customer count
- Lost customer count
- At-Risk revenue
- Lost revenue
- Top At-Risk customers by monetary value
- Average recency by segment

Business Insights

Insight 1: Champions drive the majority of revenue

Champions represent only 21.72% of customers, but generate 64.56% of total revenue, approximately 5.74M.

This demonstrates a strong concentration of revenue among highly engaged customers.

Insight 2: The business has a significant at-risk customer base

There are 661 At-Risk customers, representing 15.24% of the customer base and approximately 824K in revenue.

These customers have historically generated meaningful value but have not purchased recently.

Insight 3: Lost Customers represent the largest customer segment

Lost Customers account for 1,074 customers, or 24.76% of the customer base.

Their average recency is approximately 217 days, while their average purchase frequency is only 1.10 orders.

This indicates a substantial inactive customer population.

Insight 4: Revenue is highly concentrated

The Champions segment generates almost two-thirds of total revenue while representing roughly one-fifth of customers.

This means that changes in the behavior of high-value customers could have a significant impact on overall revenue.

Insight 5: Potential Loyalists provide a growth opportunity

Potential Loyalists represent 511 customers and have an average recency of only 16.41 days, indicating recent engagement.

However, their average frequency is only 2.12 orders.

This suggests an opportunity to encourage these customers to purchase more frequently and potentially develop stronger loyalty.

Insight 6: High-value At-Risk customers require attention

Some At-Risk customers have generated substantial historical revenue.

For example:

- Customer 15749 has monetary value of approximately 44.5K.
- Customer 15098 has monetary value of approximately 39.9K.

Although these customers are currently inactive or less engaged, their historical value makes them relevant targets for targeted retention activity.

Recommendations

Customer Segment	Recommended Action
Champions	Maintain loyalty through VIP benefits, personalized offers, early access, and exclusive rewards.
Loyal Customers	Increase purchase frequency through cross-selling, personalized recommendations, and loyalty incentives.
At-Risk Customers	Launch targeted win-back campaigns based on previous purchases and customer value.
Potential Loyalists	Encourage repeat purchases using personalized recommendations and limited-time incentives.
Lost Customers	Use selective reactivation campaigns, prioritizing customers with higher historical monetary value.
Other Customers	Use targeted engagement campaigns to identify customers with potential to move into higher-value segments.

Priority Strategy

The retention strategy should focus on two areas:

1. Protect high-value customers

Champions generate 64.56% of total revenue, so maintaining their engagement is important for revenue stability.

2. Recover valuable at-risk customers

The At-Risk segment represents approximately 824K in revenue. Customers with high historical monetary value should receive targeted win-back campaigns rather than treating all At-Risk customers identically.

Final Conclusion

The RFM analysis provides a structured view of customer purchasing behavior and highlights significant differences in customer value and engagement.

The analysis shows that Champions represent 21.72% of customers but generate 64.56% of total revenue, making them an important customer group for retention and loyalty initiatives.

At the same time, 661 customers are classified as At-Risk, representing approximately 824K in revenue, creating an opportunity for targeted win-back campaigns.

The 511 Potential Loyalists also represent a growth opportunity because they demonstrate recent purchasing activity but relatively low purchase frequency.

Overall, the project demonstrates how transactional retail data can be transformed into customer-level RFM metrics, behavioral segments, and actionable business strategies. The resulting analysis can support customer retention, loyalty development, targeted marketing, and revenue-focused customer management.